

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/372836696>

Brain tumor segmentation and classification with hybrid clustering, probabilistic neural networks

Article in *Journal of Intelligent & Fuzzy Systems* · July 2023

DOI: 10.3233/JIFS-232493

CITATIONS

31

READS

50

8 authors, including:



Javeed Md

Sree Dattha Group of Educational Institutions

38 PUBLICATIONS 156 CITATIONS

SEE PROFILE



Praveen Tumuluru

K L University CSE

57 PUBLICATIONS 440 CITATIONS

SEE PROFILE



Govinda Rajulu Ganiga

VEL TECH

12 PUBLICATIONS 37 CITATIONS

SEE PROFILE



Sanjay Kumar Suman

St. Martin's Engineering College,

44 PUBLICATIONS 406 CITATIONS

SEE PROFILE

Brain tumor segmentation and classification with hybrid clustering, probabilistic neural networks

M.D. Javeed^a, Regonda Nagaraju^b, Raja Chandrasekaran^c, Govinda Rajulu^d,
Praveen Tumuluru^e, M. Ramesh^f, Sanjay Kumar Suman^{g,*} and Rajeiv Shrivastava^h

^aDepartment of ECE and Director, IQAC, Princeton Institute of Engineering and Technology for Women, Hyderabad, Telangana, India

^bDepartment of CSE – AI&ML, School of Engineering, Mallareddy University, Hyderabad, Telangana, India

^cDepartment of ECE, Vel Tech Rangarajan Dr. Sagunthala R & D Institute of Science and Technology, Chennai, Tamil Nadu, India

^dDepartment of Computer Science and Design, St Martins Engineering college, Secunderabad, Telangana, India

^eDepartment of CSE, Koneru Lakshmaiah Education Foundation, Vaddeswaram, AP, India

^fDepartment of CSE, National Institute of Technology, Goa, India

^gDepartment of ECE, and Dean R&D, St. Martin's Engineering College Secunderabad Telangana India

^hPrinceton Institute of Engineering and Technology for Women, Hyderabad, Telangana, India

Abstract. The process of partitioning into different objects of an image is segmentation. In different major fields like face tracking, Satellite, Object Identification, Remote Sensing and majorly in medical field segmentation process is very important to find the different objects in the image. To investigate the functions and processes of human body in radiology magnetic resonance imaging (MRI) will be used. MRI technique is using in many hospitals for the diagnosis purpose widely in finding the stage of a particular disease. In this paper, we proposed a new method for detecting the tumor with enhanced performance over traditional techniques such as K-Means Clustering, fuzzy c means (FCM). Different research methods have been proposed by researchers to detect the tumor in brain. To classify normal and abnormal form of brain, a system for screening is discussed in this paper which is developed with a framework of artificial intelligence with deep learning probabilistic neural networks by focusing on hybrid clustering for segmentation on brain image and crystal contrast enhancement. Feature's extraction and classification are included in the developing process. Performance in Simulation of proposed design has shown the superior results than the traditional methods.

Keywords: Segmentation, brain tumor, probabilistic neural networks, feature extraction, classification

1. Introduction

In the investigation and to the exact disease progress in human body the Magnetic resonance imaging (MRI) [1] is used. To create these images

radio waves and Magnetic fields will be used by the help of electronic equipment. This type of practice has been using widely in hospitals for medical analysis, disease staging, and follow-up without exposing patients to ionising radiation. MRI has a wide range of applications in medical diagnosis. In treatment of many special diseases it has an influence on diagnosis even though its impact on improved health outcomes is unidentified. MRT is preferred over computed

*Corresponding author. Sanjay Kumar Suman, Professor, Department of ECE, and Dean R&D, St. Martin's Engineering College Secunderabad Telangana India. Tel.: +919652141557; E-mail: javeed.rahmanee@gmail.com.

tomography (CT) because it does not use ionising radiation and can provide the same information. The healthcare industry's sustained increase in demand for MRI has raised concerns about cost effectiveness and overdiagnosis. It takes time and effort to segment an image. Grouping similar colours or elements together is the goal of segmenting an image. If you want to do this, you can use the technique of clustering. This is done by grouping images into groups based on the similarity of colour and grey intensity.

The primary goal of clustering photographs is to extract the prominent colours from the images in question. Image segmentation may be extremely useful in simplifying complex situations by extracting information from photos such as texture, colour, shape, and structure, among other things. For the reason that it allows for the extraction of information from any image, segmentation has been applied in a variety of domains including image enhancement and compression, retrieval systems (i.e. search engines), object recognition, and medical image processing [2].

There have been numerous ways to image segmentation throughout the past few decades. Another prominent clustering method is fuzzy c-means (FCM), which uses the membership function to partition a picture into many portions [4] and [5]. To lower the computational complexity of FCM, the K-means algorithm has been devised. K-means has been widely employed in various applications because of its speed in clustering large amounts of data [4, 7, 8 and 9]. When it comes to picture segmentation, the Hierarchical clustering has also become a popular method [12, 13, and 14]. Images were then segmented using the Gaussian Mixture Model and its variation Expectation Maximization [17] and [18].

Agriculture [19] and health [20, 21] are two fields where machine learning has been extensively used for disease detection, prediction, and categorization. As far as breast cancer segmentation and classification [22–25], brain tumour detection and segmentation [26], and lung/colon cancer segmentation/classification are concerned, these are the most researched areas in health sectors.

Neurosurgical resection and pathological evaluation utilising several cellular (histologic) techniques is the gold standard for determining the presence of a brain tumour. As a result, a biopsy is an invasive procedure that can lead to bleeding and even damage, which can lead to functional loss [27]. It is because of this that non-invasive brain tumour diagnosis utilising magnetic resonance imaging (MRI) is today's mainstay of modern neuroimaging. This allows physicians

to assess the tumor's structural (cellular), metabolic (metabolic), and functional features [27, 28].

White matter (WM), grey matter (GM), and cerebrospinal fluid (CSF) are all seen in a normal structural MRI scan [29]. Water content is the primary factor that affects the appearance of these tissues when they are scanned with a structural MRI. WM, which is 70% water, is a myelinated axon that links the cerebral cortex to other brain areas. As a result, it connects the right and left hemispheres of the brain and transports information between neurons. Neuronal and glial cells drive brain activity in the grey matter, which is 80 percent water, and the basal nuclei, which lie deep within the white matter. Unlike cerebrospinal fluid, which is almost all water, cerebrospinal fluid covers all of the spaces between the brain, the skull, and the brain's ventricular system [30, 31].

It is difficult to understand the clinical manifestation of a brain tumour because of the wide range in size, location, rate of growth, and pathology of these tumours. An abnormal lump of tissue called a brain tumour is formed when some cells expand uncontrollably. Uncontrollable development in the skull interferes with normal brain activity and causes harm to the brain cells [34]. Toxic damage may occur as a result of increased brain pressure, shifts or strain on the skull, or invasion of healthy brain tissue [32, 33]. The classification of a brain tumour can be based on a variety of factors. The WHO has recommended a layered-based tumour classification schema that is more relevant to radiological applications. From top to bottom, the four layers in this diagram are: final integrated diagnosis, histologic classification, WHO grade, and molecular information. However, primary and secondary (metastatic) tumours of the brain can be categorised as such based on where they originated [35]. Cell types from which primary brain tumours are formed have been given a specific name. Non-cancerous (non-cancerous) and cancerous initial tumours are also possible (cancerous). The growth of benign tumours is slow and does not spread to other parts of the body. They can, however, place undue strain on the brain and impair its performance. Malignant tumours, on the other hand, grow quickly and spread to other tissues. Secondary brain tumours, on the other hand, originate elsewhere in the body. In most cases, these tumours are the result of cancer cells originating elsewhere in the patient's body spreading to the brain. Tumors arising in or from the lungs or breasts or melanoma or any of the above-mentioned organs are the most common primary sources of sub-

sequent brain tumour formation [31, 34, 35]. There are distinct clinical, radiological, and biochemical aspects to each of these malignancies.

Brain scans might be normal or abnormal in MRI scanning. Normal brain tissues on MRI include grey matter, white matter, and cerebrospinal fluid (CSF). The tumorous brain scan often comprises necrosis, edoema, and core tumour, in addition to the normal tissues indicated before. Inside the heart of a tumour, necrosis is a dead cell, while edoema occurs at the tumor's edges. A tumor's surrounding fluids cause edoema, which is a swelling. Non-infiltrative extra-axial cancers like meningioma can be vasogenic, whereas infiltrative tumours, such glioma, can infiltrate WM tracts of a brain [28, 36]. Structured MRI sequences such as T1-w, WT2, and FLAIR are often unable to identify these tissues from one another. Some examples of this include the difficulty in distinguishing between a tumour and its accompanying inflammation. In addition, Alves et al. [37] revealed the difficulties of distinguishing cancers based solely on signal intensity. Two patients were identified with different types of brain tumours based on similar intensity features and significant edoema around both tumours.

It is estimated that there are more than 150 different forms of central nervous system (CNS) cancers, which are usually divided into primary and metastatic (secondary) tumours. The main tumours arise in the brain or in the immediate vicinity of the brain. Metastatic tumours, on the other hand, originate elsewhere in the body and travel to the brain via the bloodstream. In contrast to primary tumours, metastatic tumours are termed cancerous or malignant.

A biopsy is the current standard approach for determining whether a brain tumour is benign or malignant. In most cases, a brain sample can only be taken through a surgical procedure that is deemed to be final. MRI-based brain tumour classification, on the other hand, does not necessitate the collection of tumour samples and is therefore less invasive. MRI-based brain tumour categorization using machine learning can also improve diagnosis and therapy planning [40]. There has been a lot of interest in developing a machine or deep learning-based system that can automatically classify brain tumours from MRI scans [39, 41–45].

An important aspect of machine learning is that it allows a machine to perform better with practise. supervised, unsupervised, and reinforcement learning are the three most prevalent forms of machine

learning approaches [46]. Unsupervised learning, on the other hand, relies on discovering patterns in data that have not been tagged by experts. Reinforcement learning, on the other hand, involves making a series of decisions based on rewards. A reward or a penalty for the actions the algorithm takes is a form of learning [46]. From MRI scans of brain tumours, machine learning has been utilised to classify the tumours. This has resulted in promising classification performance.

Preprocessing, segmentation, feature extraction, and classification are all common processes in the classic machine learning-based classification of brain tumours.

Noises, such as salt and pepper, Gaussian, Rician, and speckle noise, have a considerable impact on brain MRI scans [55–57]. In machine learning-based applications, these disturbances provide difficulties. As a result, image denoising is an essential part of the pre-processing process. Each method of MRI denoising has its own pros and limitations, which are summarised here. Statistical property and frequency spectrum distribution have been used to design a number of noise reduction strategies [58]. For example, operations such as denoise and tag removal are part of preprocessing. Also included are tasks such as smoothing the foreground and correcting intensity inhomogeneity.

It is possible to identify the tissue type and anatomical structure of an image's voxels using MRI brain scan segmentation tasks [58]. An MRI scan can be used to find the tumour area, improve visibility, and allow quantitative assessments of image structures in the feature extraction stage [47, 51]. [60] Segmentation of a brain tumour can be done in three ways: manually, semi-automatically and totally automatically.

The mathematical models used to extract features from images are based on a variety of visual characteristics. These include texture, brightness and contrast as well as Gabor transformations, gray-level co-occurrence matrix (GLCM), histogram of local binary patterns (LBP), and wavelet-based features [54, 61, 62]. CNN deep features have recently been employed as input for the SVM classifier in order to categorise brain cancers [63]. In the categorization of brain tumours, it is common practise to combine numerous variables from different extraction models [64]. For dimensionality reduction, feature selection is used.

Many authors have come up with a variety of ways to classify brain tumours based on pho-

tographs. It is possible to classify tumours in various ways, such as meningioma, glioma, and pituitary; glioma tumour grades (I–IV) [52]; benign and malignant stages (I–IV) [65–67]; diffuse midline glioma, medulloblastoma, pilocytic astrocytoma, and ependymoma; ependymoma and pilocytic astrocytomas [59]; gliomatosis and ependymoma [69].

In brain tumour classification, the most commonly used classifiers are neural networks [47–50, 70], support vector machines (SVM) [47, 54, 63, 66–69, 71, 72], K-nearest neighbour (KNN) [51, 60, 69, 63], Adaboost [65] and hybrid models [52, 74, 75]. These classifiers are all based on the neural network technology, which has been extensively studied in this field. Feedforward neural network, multilayer perceptron neural network, and probabilistic neural network (PNN) were some of the architectures used to create the neural network. There are three kernels usually used for SVM implementation: linear, homogeneous polynomial and Gaussian RBF [47–54]. That's because in the KNN classifier, a testing feature vector is categorised by finding the k-nearest training neighbour, which means that the classifier doesn't require any model to match and is just built on memory. It is important to note that the KNN algorithm uses metrics such as euclidean, city block and cosine to identify the closest distance between the testing and training class feature vectors [73, 76, 77].

Classifying brain cancers from an MRI brain image has made significant progress, but there are still significant problems in doing so using shallow supervised machine learning techniques. When it comes to obtaining descriptive information, typical hand-made feature extraction methods are inefficient [61]. The complexity of brain anatomy and the enormous density of the brain are the primary causes of this inefficiency.

Deep learning, on the other hand, is based on learning data representations and hierarchical feature learning rather than shallow machine learning techniques. Using deep learning, models for classifying brain cancers find descriptions that best describe various types of brain tumours. When it comes to brain tumour diagnosis, this type of deep learning shifts the problem from a feature-driven one to a data-driven one. For brain tumour classification, a convolutional neural network (CNN) is the most used deep learning model [39, 40].

There are a variety of classification methods for brain tumours in the reviewed literature. Pre-processing and data augmentation methods, ROI segmentation as a pre-classification phase, and

whether a pre-trained or customised deep learning model is employed are all factors that affect the final classification result. Other factors include the dataset utilised and the types of tumours included in it.

If, for example, Bada and Barjaktarovi [39] wanted to use publicly available T1-weighted brain tumour MRI scans, they could have done so. Three anatom-

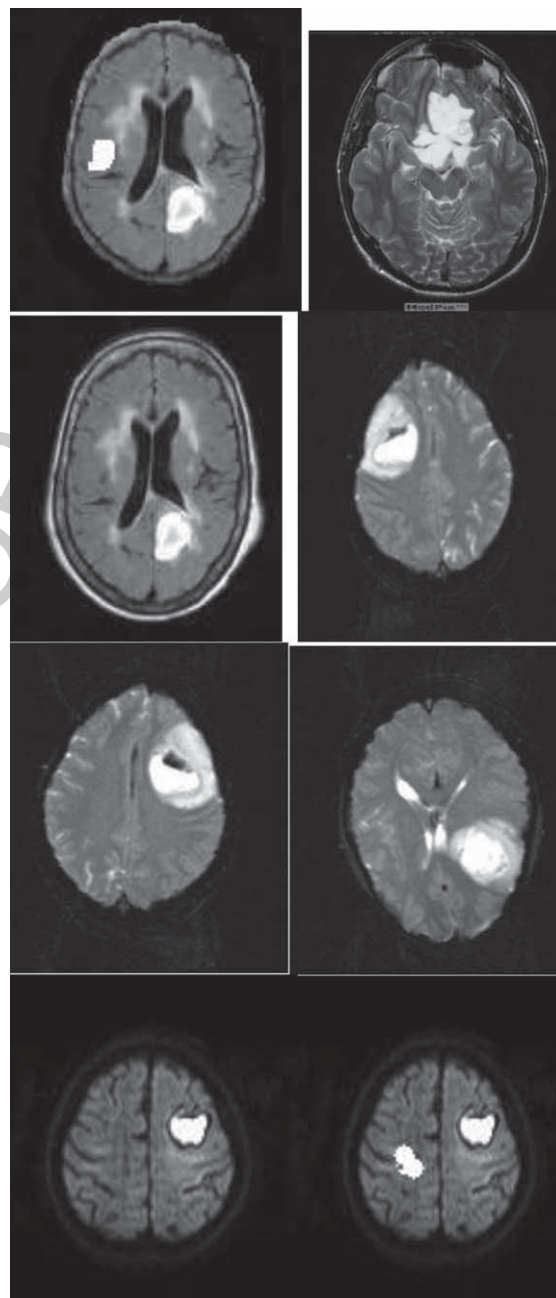


Fig. 1. Brain tumor image database.

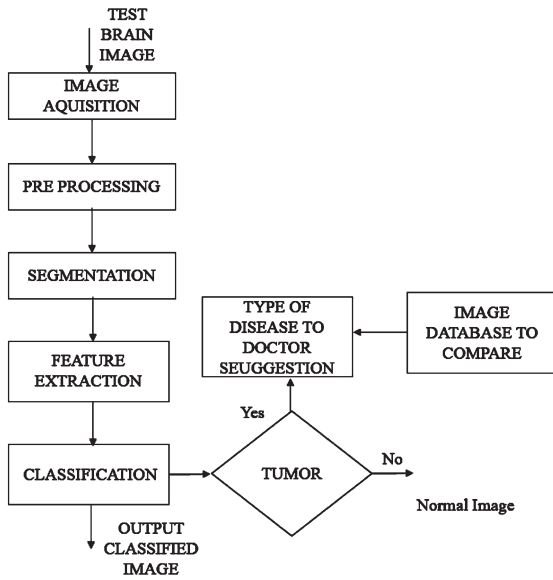


Fig. 2. Proposed structure algorithm block diagram.

ical perspectives of the brain tumours are included in this dataset: axial, transverse, and coronal sagittal. The photos were normalised and resized before they were used in the study. Adding a 90o rotation and vertical flipping to photos in the collection also increases the training dataset. The researchers employed a custom-designed CNN model trained using the Adam optimizer with a mini-batch size of 16 and tested using a 10-fold cross-validation. Initialization of the weights of convolution layers is done using a Glorot initializer. Sophisticated metrics were used to evaluate the model's performance, including sensitivity, specificity accuracy, precision recall and F1-score. As for meningioma, it's 89.8 percent, glioma 96.2 percent, and pituitary 94.4 percent. The model's specificity for meningioma, glioma, and pituitary tumours is 90.2 percent, 95.5 percent, and 96.7 percent, respectively. This study has an over-

all accuracy of 95.4 percent as well as an average precision, a mean recall and an F1-score of 95.07% [78]. This paper is organized as section 2 Discusses about the preprocessing, Section 3 about segmentation, 4 Discusses about feature extraction algorithm, Section 5 classification algorithm and 6 concludes the paper.

2. Pre-processing

The image from acquired images will be considered for further step of pre-processing, which includes background information and noise. To remove this needless information in the image, it is need to apply noise removing techniques before processing further. It is used to eliminate irrelevant information which includes noise, unwanted background part, pectoral muscle on the skin image and other artifacts. The noise of different types occurred in the skin images are Salt and Pepper, Gaussian, and Speckle and Poisson noise [8]. The different intensity will be shown by the image, whenever noise occurred in the image instead of actual pixel values of the image. Therefore, it is compulsory to choose and apply a filtering method to remove the noise as first stage. The system here has applied median filtering technique as shown in Fig. 5 and median filtered output images are in Fig. 8, which eradicates the noise from the image. These filters are capable of effectively detecting and removing image noise and fine hairs. Later, Bi Histogram equalisation works on the entire image and enhances the image, whereas adaptive histogram equalisation divides the entire image into small regions called tiles and works on each of these tiles individually. Each tile is typically 8*8 pixels and within each tile, histogram is equalized and thus it enhances the edges of the lesion. Contrast limiting is applied to limit the contrast, a level

124	112	116	127	112	115	114	117
112	110	113	118	112	119	120	121
119	117	115	119	120	127	116	127
112	116	123	124	125	118	113	118
110	113	126	127	150	116	114	116
117	114	116	112	116	127	116	127
110	113	118	110	113	118	113	118
117	114	116	117	114	116	114	116

Values 115 119 120 123
124 125 126 127 150
Median Value is 124

Fig. 3. Median filter.

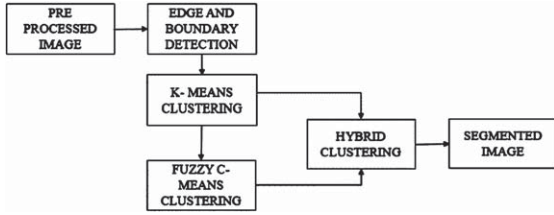


Fig. 4. Segmentation with hybrid clustering method.

below the specific limit to limit the noise. Bi-histogram equalization is applied which is focused on the average input image intensity threshold [9–10] by following the calculation of the brightness of mean and the pixels are separated into classes or sub-vector images dependent on the value of mean [8, 11].

$$\begin{aligned} I_L &= \{I(u, v) | (u, v) < I_m\}, \\ I_H &= \{I(u, v) | (u, v) \geq I_m\} \end{aligned} \quad (1)$$

Besides, the two groups create two cumulative density functions, that is,

$$\begin{aligned} C_L(i) &= \sum_{j=0}^i h(j), \quad i = 0, \dots, I_m - 1. \\ C_H(j) &= \sum_{k=I_m}^j h(k), \quad j = I_m, \dots, L - 1 \end{aligned} \quad (2)$$

Where $\sum i^c L(i) = \sum j^c H(j) = 1$ each sub-image is processed to provide improved contrast performance,

$$\begin{aligned} I_{L, enh}(i) &= (I_m - 1) \times C_L(i) \\ I_{H, enh}(j) &= I_m + (L - 1 - I_m) \times C_H(j) \end{aligned} \quad (3)$$

From the set of sub-images, the output and enhanced image are obtained.

$$I_{enh} = I_{L, enh} \cup I_{H, enh}. \quad (4)$$

The average image brightness is the average value of the average luminosity of sub-images. The output is differed from the average input brightness is the changed histogram of crystal contrast enhanced images to input images.

3. Segmentation

After the pre-processing stage, segmentation of lesion was carried out to get the transparent portion

of the affected area of skin. On performing transformation, Hybrid clustering technique is applied on the processed image to segment the skin lesion area based on Fuzzy C-Means clustering and K-Means techniques. In K-means technique, Segmentation is the preliminary process of this technique, at the cluster centres, cost junction must be minimized which randomly varies with respect to memberships of user inputs. Dividing an image into multiple clusters is the main function of segmentation and this is based on region of interest to identify the skin cancer. As described in Fig. 3, In this proposed system, K-Means clustering technique is applied first after detecting the specific edges and later fuzzy c means clustering is applied.

3.1. K-Means clustering

The key structure of K-Means Clustering technique is discussed in this section. Let $Y = \{y_i | i = 1, \dots, P\}$ be features of P-dimensional vectors and $Z = \{z_j | j = 1, \dots, N\}$ be each data of Y. K-means clusters which Z is $TP = \{T_j | 1, \dots, k\}$. The image is separated into clusters and every cluster will have average centre value. K-Means clustering contains the subsequent phases as discussed below

1. At first, generate with centroids C, the random starting points.
2. Separation between pixel and centre has to discover by Euclidean distance i.e. from X to C.
3. From the partition of X_i for $i = 1 \dots N$ Determine the base $d(z_i, C)$.
4. Discovering cluster centre which is new C_i for $i = 1 \dots k$ is defined as:

$$C_i = \frac{1}{n_i} \sum_{j=1}^{n(si)} m_{ij} \in si \quad (5)$$

5. The procedure need be repeated from step 2 for all the clusters.

The centroids are said to be converged if they are not changing their position and they may stop in any particular cluster 't' with a threshold TH, then the positions have to be refreshed by considering the threshold value TH. The Fig. 4 represents the step-by-step procedure for K-Means clustering.

$$\left| \frac{c^t - c^{t-1}}{c^t} \right| < TH \quad (6)$$

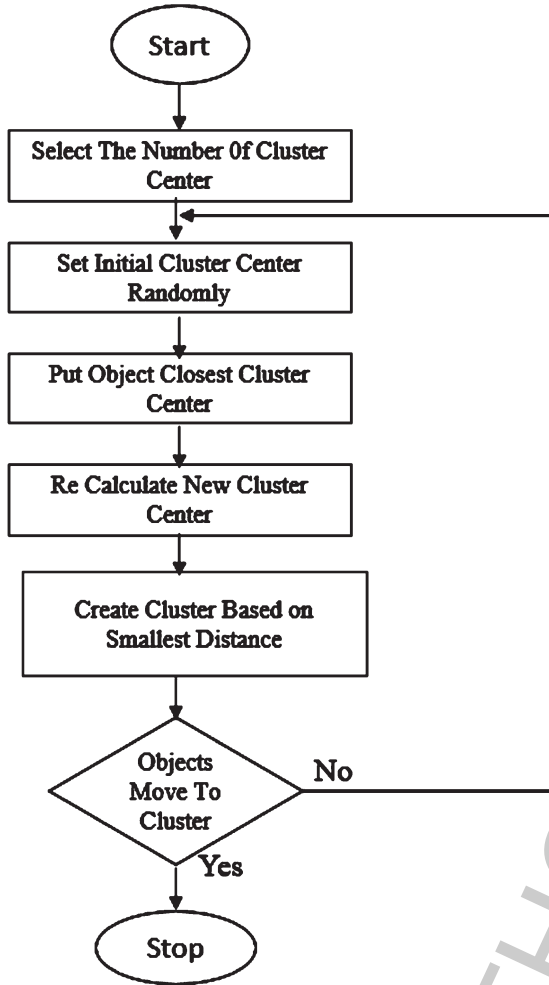


Fig. 5. K-Means clustering procedure.

3.2. Fuzzy C-means clustering

In general, segmentation fuzzy c-mean algorithm is one which will be used for dividing the images with space of image to numerous cluster areas with parallel image pixel values. It is very useful for medical image analysis, as it is fuzzified type of the k-means segmentation technique. The Fuzzy C-Means (FCM) clustering algorithm introduced first in [12] and then it was improved in [13]. The procedure is an iterative clustering technique which produces an ideal C partition by lessening the weighted inside group sum of squared error and is extensively used in pattern recognition and segmentation of images. This algorithm will have the following steps traditionally.

$$\forall x \left(\sum_{k=1}^{num.clusters} u_k(x) = 1 \right) \quad (7)$$

The distance between the cluster to centre in relation with above equations is and the algorithm centroids with a cluster of all the points is shown as

$$Center_k = \frac{\sum_x u_k(x)^m x}{\sum_x u_k(x)^m} \quad (8)$$

$$u_k(x) = \frac{1}{d(center_k, x)} \quad (9)$$

Then coefficient is a proper parameter to proper distribution >1 So there is 1.

$$u_k(x) = \frac{1}{\sum_j \left(\frac{d(center_k, x)}{d(center_j, x)} \right)^{2/(m-1)}} \quad (10)$$

The equivalent of 2 m for a linear normalization of coefficients equalling their quantity to 1. When 1 m is closer, the cluster is much weighted more than other clusters at this stage and is similar to the K-means algorithm [14].

Fuzzy C-Means clustering technique looks almost similar to K-Means clustering and contains the subsequent phases as discussed below

- Need to select the cluster randomly in number
- Repeat the algorithm as discussed in each step of equations 7,8,9,10
- Calculate the centre using the above equation 8
- Calculate the coefficients by using center value from 8 using equations 9 and 10.

4. Feature extraction

From the basic and raw information, extracting of required image data is important and it should be most relevant for further procedure like classification. The information dimensionality of the data may be reduced while extracting the features. It is done here due to technical limits in time for computation and memory. A normal and quality feature extraction technique should have the capability to enhance and uphold the features of input image raw data which makes difference patterns from each other. At the same time, while acquiring the image, it should be capable to process by human being. The major features are extracted: Texture features using Gray Level Cooccurrence Matrix (GLCM), Low level features using Discrete Wavelet Transform (DWT), Colour features. These are the most common features used for the extraction of features, though the performance of three techniques dependent on the image raw data.

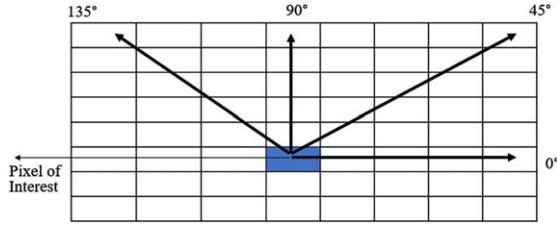


Fig. 6. Directional analysis of GLCM.

To continue the developments in image processing, computer vision extraction of features is having a large role as it has a rich history of fields of classifying images, medical imaging, remote sensing, visual human perceptions. Word texture depends always on human perception as interpretations of individual will vary based on that particular texture. It provides a meaningful result as a quantify we can define it as texture. In an image, pivotal component is always texture and it uses to explain about spatial differences in a raw segmented image at particular point, to define brightness, regularity and coarseness of the image. Especially in diagnosis analysis of texture, it is very important to give a solution to a problem by performing manipulations on digital images which can be easy for a human expert eye. The system here extracts three features as discussed earlier. First, the GLCM features are given importance and extracted in the proposed work: Gray levels, Energy, Contrast, Inverse Difference and entropy. In [15] Haralick et al have proposed different texture features of characteristics complexity information. It has become wide for many applications since the authors proposed their own work features. Moreover, GLCM characteristic features with image enhancement focus was discussed in [16]. The procedure of GLCM is explained in following figures. Different four directions of GLCM for the same image or image sub-regions are shown in the Fig. 5. Figure 6 shows the test image and Fig. 7 is the general structure of GLCM, Fig. 8(a), (b), (c), (d) are GLCM with relative distance 'δ' and with different inclinations.

4.1. Texture feature based on GLCM

Analysis of texture will concentrate on qualities of an image like smoothness, silky, rough or bumpy as described as function with spatial variations in gray levels. GLCM is a regularity matrix by which its level is known by image gray level values. GLCM produces a matrix with distance between pixels and

1	0	1	2
2	0	1	3
1	2	3	1
1	1	0	3

Test Image

m/n	0	1	2	3
0	#(0,0)	#(0,0)	#(0,0)	#(0,0)
1	#(0,0)	#(0,0)	#(0,0)	#(0,0)
2	#(0,0)	#(0,0)	#(0,0)	#(0,0)
3	#(0,0)	#(0,0)	#(0,0)	#(0,0)

Fig. 7. General form of GLCM.

0	2	0	0
1	0	2	1
1	1	0	0
0	0	0	1

(a)

0	2	0	1
1	1	2	1
1	0	0	1
0	1	0	0

(b)

1	0	0	1
0	2	2	1
1	1	0	0
0	2	1	0

(c)

0	1	1	0
1	2	0	0
0	0	1	0
1	1	0	1

(d)

Fig. 8. GLCM for (a) $\delta = 1, \theta = 135^\circ$ (b) $\delta = 1, \theta = 45^\circ$ (c) $\delta = 1, \theta = 90^\circ$ (d) $\delta = 1, \theta = 0^\circ$.

different inclinations as shown in Figs. 5, 6, 7 and to extract as texture features with eloquent statistics from the generated matrix. To compose GLCM, main property is probability value which can be defined as $P(m, n | \delta, \theta)$ to express the probability of pixels at θ inclination and δ interval. $P(m, n | \delta, \theta)$ is represented by $P(m, n)$ when θ and δ are known [17, 18]. The equation for the Elements to compute is

$$P(m, n | \delta, \theta) = \frac{P(m, n | \delta, \theta)}{\sum_m \sum_n P(m, n | \delta, \theta)} \quad (11)$$

Usually, GLCM describes all the texture features, however here the system mainly focusses on entropy, energy, inverse difference and contrast as features.

A. Entropy

Randomness of image texture while co-occurrence matrix is equal for all the values and the maximum entropy depends on random distribution of gray levels as

$$S = - \sum_x \sum_y p(x, y) \log p(x, y) \quad (12)$$

B. Energy

Homogeneity texture measures the changes in reflection while distribution gray level uniformity in

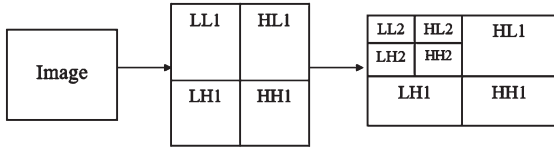


Fig. 9. Level DWT coefficients.

the image as

$$E = \sum_m \sum_n P(x, y)^2 \quad (13)$$

C. Inverse difference

Changes in texture image number is calculated by inverse difference. While $p(x, y)$ considering as gray level at a particular point and co-ordinate (x, y) inverse difference is as

$$H = \sum_x \sum_y \frac{1}{1 + (x - y)^2} P(x, y) \quad (14)$$

D. Contrast

Local number changes with the value distributed which reflects in image clarity and shadow depth can be calculated by Contrast as

$$I = \sum \sum (x - y)^2 p(x, y) \quad (15)$$

Then, level 2 DWT is also used to extract the low-level features and initially on the segmented output D is applied, it yields the outputs as LL1, LH1, HL1 and HH1 bands respectively. Then entropy, energy and correlation features are calculated on the LL band. Then, on the LL output band again DWT is applied, and it provides the outputs as LL2, LH2, HL2 and HH2 respectively. Again entropy, energy and correlation features are calculated on the LL2 band respectively as shown in figure.

And finally, Mean, and standard deviation based Statistical Colour features are extracted from the segmented image and specified in the below equations,

$$\text{Mean } (\mu) = \frac{1}{N^2} \sum_{i,j=1}^N I(i, j) \quad (16)$$

$$\text{Standard Deviation } (\sigma) = \sqrt{\frac{\sum_{i,j=1}^N [I(i, j) - \mu]^2}{N^2}} \quad (17)$$

Later, all these features are combined using array concatenation and it produces the output as hybrid feature matrix.

5. Classification

For example, in finance, medicine, engineering, geology, physics and biology neural networks have been used successfully to solve a variety of problems. As a means of solving problems involving categorization and prediction, neural networks have much promise from a statistical perspective. An emulation of the birth neural system is used to build PNN. The neurons are wired together in a predetermined way so that categorization may be carried out efficiently. The weights of the neurons are determined by the hybrid properties of the system. The hybrid features of the system are then used to identify the relationships between weights. In the proposed network, the number of weights determines how many layers there will be. Figure 9 shows the architecture of artificial neural networks. Training and testing are the two main stages of PNN categorization. The layer-based architecture will be used for training purposes. Feature weight distributions are mapped onto the input dataset using the input layer; this dataset is classed into hybrid features.

Weights are assigned to each of the PNN's four hidden layers. Prior to class node activation and decision normalisation layers, an initial convolutional2D hidden layer of the network processes images of skin lesions that measure 224*224*3 pixels. Then, in the two tiers of class nodes concealed layer, the classification operation was carried out. The two layers of the concealed layer include information about the typical and pathological characteristics of skin cancer. Classification stages are classed into normal and pathological according to criteria. In the output layer, the labels for these two levels are mapped. Again, the second level of the hidden layer comprises the aberrant cancer types, as well as the benign and malignant cancer weights; this is the second stage of hidden

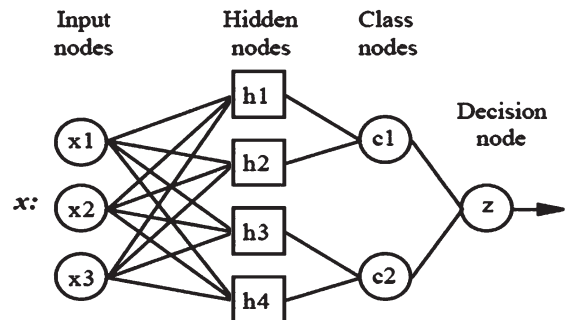


Fig. 10. Layered architecture of PNN model [79].

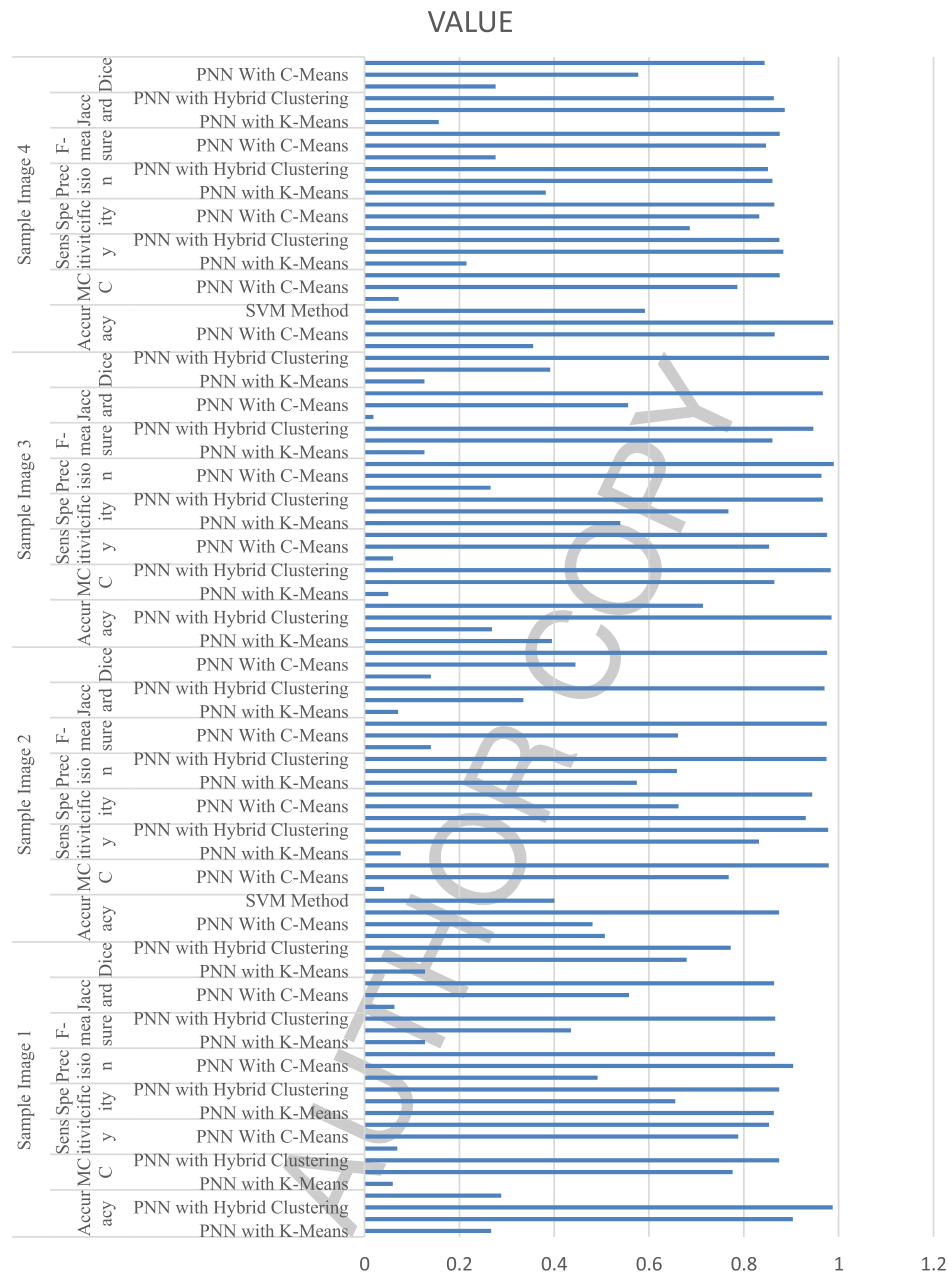
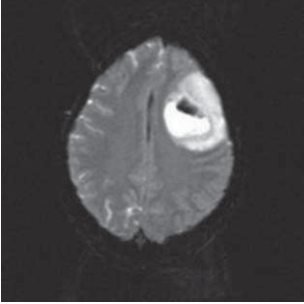
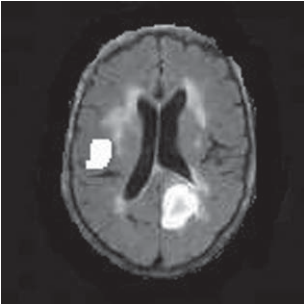


Fig. 11. Performance metrics.

layer. There is a similar mapping of these benign and malignant weights into the output layer. The hybrid features of the test picture are used for testing purposes in the classification step when the test image is applied. Using Euclidean distance as the maximum feature matching criteria, it will work. A normal skin image is defined as one in which the feature match happened with the labels on layer 1. If the feature

match occurred with the C1 labels with the greatest weight distribution, then the image is classed as benign affected cancer. If the feature match occurs with C2 labels with the smallest weight distribution, it is considered a malignant impacted cancer image. It will then be determined what type of sickness it is based on the features and classification of the patient's illness [79].

Table 1
Performance metrics

Image	Parameter	Method	Value
Sample Image 1 	Accuracy	PNN with K-Means	0.26713
		PNN With C-Means	0.90372
		PNN with Hybrid Clustering	0.98723
		SVM Method	0.28831
	MCC	PNN with K-Means	0.059602
		PNN With C-Means	0.77651
		PNN with Hybrid Clustering	0.87423
	Sensitivity	PNN with K-Means	0.069035
		PNN With C-Means	0.78821
		PNN with Hybrid Clustering	0.85315
	Specificity	PNN with K-Means	0.86331
		PNN With C-Means	0.65548
		PNN with Hybrid Clustering	0.87475
	Precision	PNN with K-Means	0.49166
		PNN With C-Means	0.90421
		PNN with Hybrid Clustering	0.8657
Sample Image 2 	Accuracy	PNN with K-Means	0.50714
		PNN With C-Means	0.48125
		PNN with Hybrid Clustering	0.87465
		SVM Method	0.4005
	MCC	PNN with K-Means	0.041112
		PNN With C-Means	0.76834
		PNN with Hybrid Clustering	0.97928
	Sensitivity	PNN with K-Means	0.075884
		PNN With C-Means	0.83221
		PNN with Hybrid Clustering	0.97792
	Specificity	PNN with K-Means	0.93062
		PNN With C-Means	0.6624
		PNN with Hybrid Clustering	0.94447
	Precision	PNN with K-Means	0.57454
		PNN With C-Means	0.65856
		PNN with Hybrid Clustering	0.97454
	F-measure	PNN with K-Means	0.13976
		PNN With C-Means	0.66121
		PNN with Hybrid Clustering	0.97512
	Jaccard	PNN with K-Means	0.070943
		PNN With C-Means	0.33492
		PNN with Hybrid Clustering	0.97054
	Dice	PNN with K-Means	0.13976
		PNN With C-Means	0.44494
		PNN with Hybrid Clustering	0.97556

(Continued)

Table 1
(Continued)

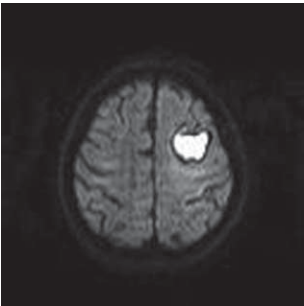
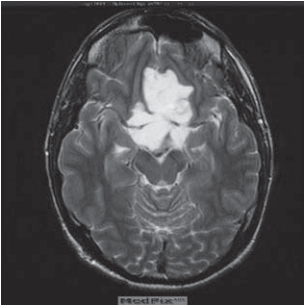
Image	Parameter	Method	Value
Sample Image 3 	Accuracy	PNN with K-Means	0.39548
		PNN With C-Means	0.26857
		PNN with Hybrid Clustering	0.98512
		SVM Method	0.7135
	MCC	PNN with K-Means	0.05021
		PNN With C-Means	0.86452
		PNN with Hybrid Clustering	0.98324
	Sensitivity	PNN with K-Means	0.06032
		PNN With C-Means	0.85321
		PNN with Hybrid Clustering	0.97562
	Specificity	PNN with K-Means	0.53945
		PNN With C-Means	0.76762
		PNN with Hybrid Clustering	0.96684
	Precision	PNN with K-Means	0.26582
		PNN With C-Means	0.96362
		PNN with Hybrid Clustering	0.98998
Sample Image 4 	Accuracy	PNN with K-Means	0.35582
		PNN With C-Means	0.86478
		PNN with Hybrid Clustering	0.98856
		SVM Method	0.5916
	MCC	PNN with K-Means	0.072041
		PNN With C-Means	0.78652
		PNN with Hybrid Clustering	0.87552
	Sensitivity	PNN with K-Means	0.21527
		PNN With C-Means	0.88325
		PNN with Hybrid Clustering	0.87531
	Specificity	PNN with K-Means	0.68616
		PNN With C-Means	0.83239
		PNN with Hybrid Clustering	0.86475
	Precision	PNN with K-Means	0.38226
		PNN With C-Means	0.86015
		PNN with Hybrid Clustering	0.85102
	F-measure	PNN with K-Means	0.27633
		PNN With C-Means	0.84677
		PNN with Hybrid Clustering	0.87575
Jaccard		PNN with K-Means	0.15645
		PNN With C-Means	0.88624
		PNN with Hybrid Clustering	0.86352
	Dice	PNN with K-Means	0.27633
		PNN With C-Means	0.577423
		PNN with Hybrid Clustering	0.84381

Table 2
CPU computation time for different segmentation techniques

S. No.	Segmentation algorithm	Computation time (Seconds)			
		Sample Image 1	Sample Image 2	Sample Image 3	Sample Image 4
1	Fuzzy C Means	11.2633	3.4008	2.8348	2.9282
2	K-Means	0.4056	0.5148	0.4572	0.3432
3	Proposed	0.2496	0.2340	0.2712	0.2798

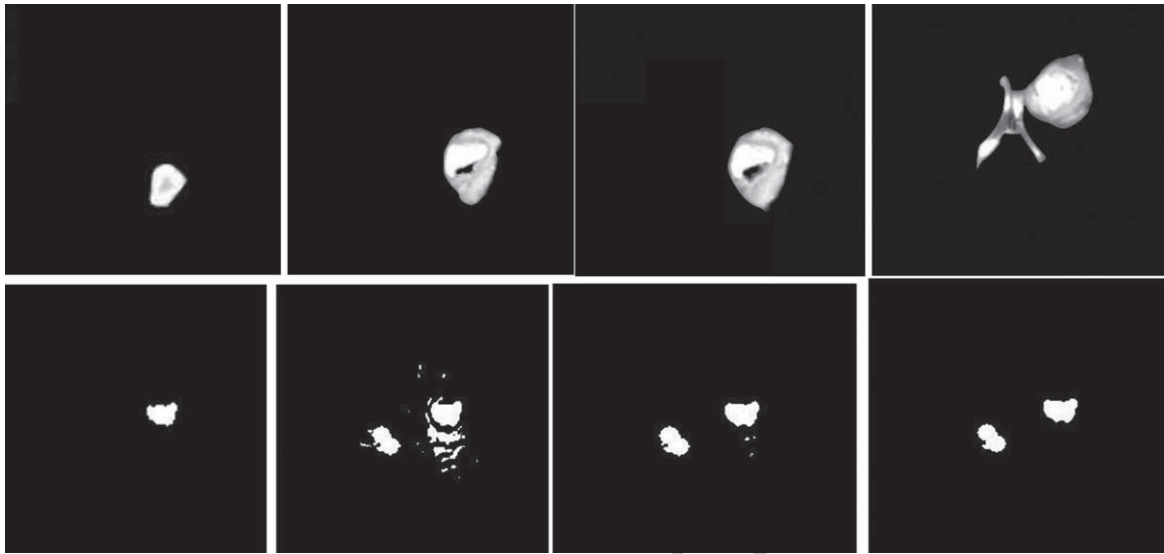


Fig. 12. Segmented images-PNN with hybrid clustering.

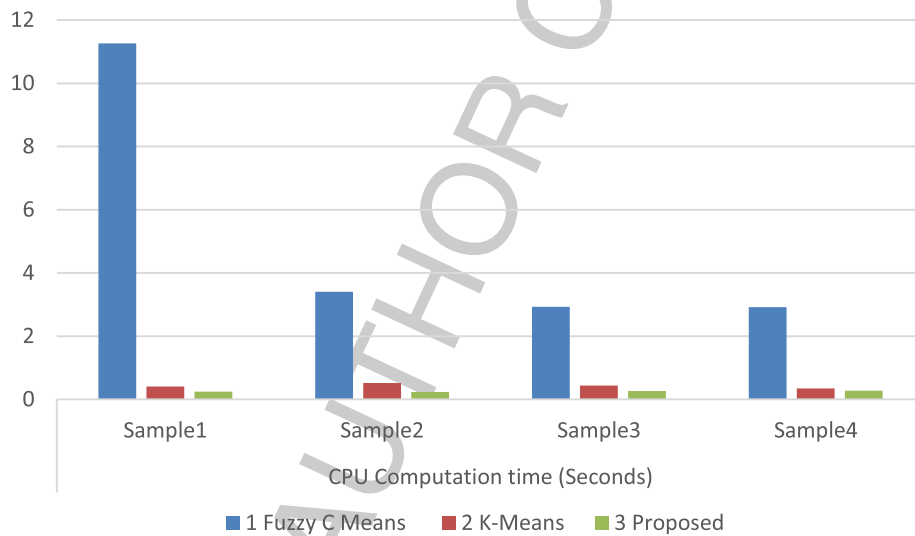


Fig. 13. CPU computation time for different segmentation techniques.

6. Conclusion

In this paper, we had presented a new MR brain image segmentation for more accurately and more quickly detecting tumours in the brain. Clustering strategy for lowering computing time and the binarization method for calculating the area in terms of mm² based on typography and digital imaging units are the subjects of this article. In order to compare the simulation results of the existing algorithms with the proposed algorithm, we calculated the CPU time. Last but not least, the new approach outperformed

the old one in terms of both computational time and precision. The future neural networks with artificial intelligence and machine learning algorithms will play a vital role in the field of medical imaging [80, 81].

References

- [1] M. Zubair, I.A. Rana, Y. Islam and S.A. Khan, Variable Structure Based Control for the Chemotherapy of Brain Tumor, in *IEEE Access* **9** (2021), 107333–107346. doi: 10.1109/ACCESS.2021.3091632

- [2] I. Khan, K. Ahsan, M.A. Hasan and A. Sattar, Brain Tumor Analysis Using Deep Neural Network, *2021 5th International Conference on Intelligent Computing and Control Systems (ICICCS)*, 2021, pp. 1749–1754, doi: 10.1109/ICICCS51141.2021.9432235
- [3] Manisha Bhagwat, R.K. Krishna and V.E. Pise, Image Segmentation by Improved Watershed Transformation in Programming Environment MATLAB, *International Journal of Computer Science & Communication* **1**(2) (2010), 171–174.
- [4] M.H. Fazel Zarandia, M. Zarinbala and M. Izadi, Systematic image processing for diagnosing brain tumours: A Type-II fuzzy expert system approach, *Applied Soft Computing*, pp: 285–294, 2011.
- [5] S. ZulaikhaBeevi, M. Mohamed Sathik, An Effective Approach for Segmentation of MRI Images: Combining Spatial Information with Fuzzy C-Means Clustering, *European Journal of Scientific Research* **41**(3) (2010), 437–451.
- [6] S. Mary Praveena, Dr. IlaVennila, Optimization Fusion Approach for Image Segmentation Using K-Means Algorithm, *International Journal of Computer Applications* **2**(7) (2010).
- [7] M. Masroor Ahmed and Dzulkifli Bin Mohammad, Segmentation of Brain MR Images for Tumour Extraction by Combining K-means Clustering and Perona-Malik Anisotropic Diffusion Model, *International Journal of Image Processing* **2**(1) (2010).
- [8] Tse-Wei Chen, Yi-Ling Chen and Shao-Yi Chien, Fast Image Segmentation Based on K-Means Clustering with Histograms in HSV Color Space, *Journal of Scientific Research* **44**(2) (2010), 337–351.
- [9] Anil Z. Chitade, Colour based image segmentation using k-means clustering, *International Journal of Engineering Science and Technology* **2**(10) (2010), 5319–5325.
- [10] J. Selvakumar, A. Lakshmi and T. Arivoli, Brain Tumour segmentation and its area Calculation in Brain MR images using K-means Clustering and Fuzzy C-means algorithm, *International Conference on Advances in Engineering, Science and Management (ICAESM)*, pp: 186–190, 2012.
- [11] A.R. Barakbah and Y. Kiyoki, A Pillar algorithm for K-means Optimization by Distance Maximization for Initial Centroid Designation, *IEEE Symposium on Computational Intelligence and Data Mining*, pp: 61–68, 2009.
- [12] A.M. Usó, F. Pla and P.G. Sevilla, Unsupervised Image Segmentation Using a Hierarchical Clustering Selection Process, *Structural, Syntactic, and Statistical Pattern Recognition* **4109** (2006), 799–807.
- [13] A.Z. Arifin and A. Asano, Image segmentation by histogram thresholding using hierarchical cluster analysis, *Pattern Recognition Letters* **27**(13) (2006), 1515–1521.
- [14] B. Micušik and A. Hanbury, Automatic Image Segmentation by Positioning a Seed*, *ECCV 2006, Part II, LNCS 3952*, Springer Berlin/Heidelberg, pp. 468–480, 2006.
- [15] J. Chen, J. Benesty, Y.A. Huang and S. Doclo, New Insights Into the Noise Reduction Wiener Filter, *IEEE Transactions on Audio, Speech, and Language Processing* **14**(4) (2006).
- [16] Y. Pan, J.D. Birdwell and S.M. Djouadi, Bottom-Up Hierarchical Image Segmentation Using Region Competition and the Mumford-Shah Functional, *Proc. 18th International Conference on Pattern Recognition (ICPR)* **2** (2006), 117–121.
- [17] C. Carson and H. Greenspan, Blob world: Image Segmentation Using Expectation-Maximization and Its Application to Image Querying, *IEEE Transactions On Pattern Analysis And Machine Intelligence* **24**(8) (2002), 1026–1038.
- [18] C.J. Veenman, M.J.T. Reinders and E. Backer, A maximum variance cluster algorithm, *IEEE Transactions On Pattern Analysis And Machine Intelligence* **24**(9) (2002), 1273–1280.
- [19] Y.K. Afework and T.G. Debelee, Detection of Bacterial Wilt on Enset Crop Using Deep Learning Approach, *Int J Eng Res Afr* **51** (2020), 131–146.
- [20] T.G. Debelee, F. Schwenker, A. Ibenthal and D. Yohannes, Survey of deep learning in breast cancer image analysis, *Evol Syst* **11** (2019), 143–163.
- [21] T.G. Debelee, S.R. Kebede, F. Schwenker and Z.M. Shewarega, Deep Learning in Selected Cancers' Image Analysis—A Survey, *J Imaging* **6** (2020), 121.
- [22] T.G. Debelee, M. Amirian, A. Ibenthal, G. Palm and F. Schwenker, Classification of Mammograms Using Convolutional Neural Network Based Feature Extraction, In *Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering*; Springer International Publishing: Berlin/Heidelberg, Germany, 2018; pp. 89–98.
- [23] T.G. Debelee, A. Gebreselasie, F. Schwenker, M. Amirian and D. Yohannes, Classification of Mammograms Using Texture and CNN Based Extracted Features, *J Biomimetics Biomater Biomed Eng* **42** (2019), 79–97.
- [24] T.G. Debelee, F. Schwenker, S. Rahimeto and D. Yohannes, Evaluation of modified adaptive k-means segmentation algorithm, *Comput Vis Media* **5** (2019), 347–361.
- [25] S.R. Kebede, T.G. Debelee, F. Schwenker and D. Yohannes, Classifier Based Breast Cancer Segmentation, *J Biomimetics Biomater Biomed Eng* **47** (2020), 41–61.
- [26] Y. Megersa and G. Alemu, Brain tumor detection and segmentation using hybrid intelligent algorithms, In *Proceedings of the AFRICON 2015*, Addis Ababa, Ethiopia, 14–17 September 2015.
- [27] T.A. Roberts, H. Hyare, G. Agliardi, B. Hipwell, A. d'Esposito, A. Ianus, J.O. Breen-Norris, R. Ramasawmy, V. Taylor, D. Atkinson, et al., Noninvasive diffusion magnetic resonance imaging of brain tumour cell size for the early detection of therapeutic response, *Sci Rep* (2020).
- [28] J.E. Villanueva-Meyer, M.C. Mabray and S. Cha, Current Clinical Brain Tumor Imaging, *Neurosurgery* **81** (2017), 397–415.
- [29] M.J. Rosenbloom and A. Pfefferbaum, Magnetic resonance imaging of the living brain: evidence for brain degeneration among alcoholics and recovery with abstinence, *Alcohol Res Health J Natl Inst Alcohol Abuse Alcohol* **31** (2008), 362–37.
- [30] Charles R. Noback, Norman L. Strominger, R.J. Demarest and D.A. Ruggiero, *The Human Nervous System: Structure and Function*; Humana Press: Totowa, NJ, USA, 2005.
- [31] D.N. Louis, H. Ohgaki and W.O., WHO Classification of Tumors of the Central Nervous System; *International Agency for Research on Cancer (IARC)*: Lyon, France, 2007.
- [32] A.A. Kayode, A. Shahzadi, M. Akram, H. Anwar, O.T. Kayode, O.O. Akinlawo and S.O. Okoh, Brain Tumor: An overview of the basic clinical manifestations and treatment, *Glob J Cancer Ther* **2020** (2020), 38–41.
- [33] D.R. Johnson, J.B. Guerin, C. Giannini, J.M. Morris, L.J. Eckel and T.J. Kaufmann, 2016 Updates to the WHO Brain Tumor Classification System: What the Radiologist Needs to Know, *RadioGraphics* **37** (2017), 2164–2180.
- [34] P. Roth, A. Pace, E.L. Rhun, M. Weller, C. Ay, E.C.J. Moyal, M. Coomans, R. Giusti, K. Jordan, N. Nishikawa, et al., Neurological and vascular complications of primary and secondary brain tumours: EANO-ESMO Clinical Practice

- Guidelines for prophylaxis, diagnosis, treatment and follow-up, *Ann Oncol* **32** (2021), 171–182.
- [35] Jan C. Buckner, e.a. Central Nervous System Tumors, In Mayo Clinic Proceedings; Elsevier: Amsterdam, The Netherlands, 2007; Volume 82, pp. 1271–1286.
- [36] R. Smithuis, Neuroradiology: Brain Index. Available online: <https://radiologyassistant.nl/neuroradiology/brain> (accessed on 3 March 2021).
- [37] A.F.F. Alves, J.R. de Arruda Miranda, F. Reis, S.A.S. de Souza, L.L.R. Alves, L. de Moura Feitoza, J.T. de Souza de Castro and D.R. de Pina, Inflammatory lesions and brain tumors: is it possible to differentiate them based on texture features in magnetic resonance imaging? *J Venom Anim Toxins Incl Trop Dis* **26** (2020).
- [38] P. Kleihues, D.N. Louis, B.W. Scheithauer, L.B. Rorke, G. Reifenberger, P.C. Burger and W.K. Cavenee, The WHO Classification of Tumors of the Nervous System, *J Neuropathol Exp Neurol* **61** (2002), 215–225.
- [39] M.M. Badža and M.C. Barjaktarović, Classification of Brain Tumors from MRI Images Using a Convolutional Neural Network, *Appl Sci* **10** (2020), 1999.
- [40] G.S. Tandel, M. Biswas, O.G. Kakde, A. Tiwari, H.S. Suri, M. Turk, J. Laird, C. Asare, A.A. Ankrah, N.N. Khanna, et al., A Review on a Deep Learning Perspective in Brain Cancer Classification, *Cancers* **11** (2019), 111.
- [41] J. Quon, W. Bala, L. Chen, J. Wright, L. Kim, M. Han, K. Shpanskaya, E. Lee, E. Tong, M. Iv, et al., Deep Learning for Pediatric Posterior Fossa Tumor Detection and Classification: A Multi-Institutional Study, *Am J Neuroradiol* (2020).
- [42] F.J. Díaz-Pernas, M. Martínez-Zarzuela, M. Antón-Rodríguez and D. González-Ortega, A Deep Learning Approach for Brain Tumor Classification and Segmentation Using a Multiscale Convolutional Neural Network, *Healthcare* **9** (2021), 153.
- [43] S. Deepak and P. Ameer, Brain tumor classification using deep CNN features via transfer learning, *Comput Biol Med* **111** (2019), 103345.
- [44] J.S. Paul, A.J. Plassard, B.A. Landman and D. Fabbri, Deep learning for brain tumor classification, In *Medical Imaging 2017: Biomedical Applications in Molecular, Structural, and Functional Imaging*; A. Krol, B. Gimi, Eds.; SPIE: Bellingham, WA, USA, 2017.
- [45] H.A. Khan, W. Jue, M. Mushtaq and M.U. Mushtaq, Brain tumor classification in MRI image using convolutional neural network, *Math Biosci Eng* **17** (2020), 6203–6216.
- [46] P. Dangeti, Statistics for Machine Learning; Packt Publishing: Birmingham, UK, 2017.
- [47] R. Ahmed, A.S. Swakshar, M.F. Hossain and M.A. Rafiq, Classification of tumors and it stages in brain MRI using support vector machine and artificial neural network, In *Proceedings of the 2017 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, Cox's Bazar, Bangladesh, 16–18 Feb. 2017, pp. 229–234.
- [48] M.R. Ismael and I. Abdel-Qader, Brain Tumor Classification via Statistical Features and Back-Propagation Neural Network, In *Proceedings of the 2018 IEEE International Conference on Electro/Information Technology (EIT)*, Rochester, MI, USA, 3–5 May 2018, pp. 0252–0257.
- [49] K.A. Sathi and M.S. Islam, Hybrid Feature Extraction Based Brain Tumor Classification using an Artificial Neural Network, In *Proceedings of the 2020 IEEE 5th International Conference on Computing Communication and Automation (ICCCA)*, Greater Noida, India, 30–31 October 2020, pp. 155–160.
- [50] N.V. Shree and T.N.R. Kumar, Identification and classification of brain tumor MRI images with feature extraction using DWT and probabilistic neural network, *Brain Inform* **5** (2018), 23–30.
- [51] R.H. Ramdlon, E.M. Kusumaningtyas and T. Karlita, Brain Tumor Classification Using MRI Images with K-Nearest Neighbor Method, In *Proceedings of the 2019 International Electronics Symposium (IES)*, Surabaya, Indonesia, 27–28 Sept. 2019, pp. 660–667.
- [52] G. Garg and R. Garg, Brain Tumor Detection and Classification based on Hybrid Ensemble Classifier, arXiv 2021, arXiv:2101.00216.
- [53] E. N., N. M. and W. Al-Atabany, Evaluating the Efficiency of different Feature Sets on Brain Tumor Classification in MR Images, *Int J Comput Appl* **180** (2018), 1–7.
- [54] M. Gurbina, M. Lascu and D. Lascu, Tumor Detection and Classification of MRI Brain Image using Different Wavelet Transforms and Support Vector Machines, In *Proceedings of the 2019 42nd International Conference on Telecommunications and Signal Processing (TSP)*, Budapest, Hungary, 1–3 July 2019, pp. 505–508.
- [55] H.M. Ali, MRI Medical Image Denoising by Fundamental Filters, In *High-Resolution Neuroimaging—Basic Physical Principles and Clinical Applications*; In Tech: Vienna, Austria, 2018.
- [56] L. Liu, H. Yang, J. Fan, R.W. Liu and Y. Duan, Rician noise and intensity nonuniformity correction (NNC) model for MRI data, *Biomed Signal Process Control* **49** (2019), 506–519.
- [57] S. Ramesh, S. Sasikala and N. Paramanandham, Segmentation and classification of brain tumors using modified median noise filter and deep learning approaches, *Multimed Tools Appl* **80** (2021), 11789–11813.
- [58] D.V.S. Ravikumar Gurusamy, A Machine Learning Approach for MRI Brain Tumor Classification, *Comput Mater Contin* **53** (2017), 91–108.
- [59] M. Li, H. Wang, Z. Shang, Z. Yang, Y. Zhang and H. Wan, Ependymoma and pilocytic astrocytoma: Differentiation using radiomics approach based on machine learning, *J Clin Neurosci* **78** (2020), 175–180.
- [60] K. Kaplan, Y. Kaya, M. Kuncan and H.M. Ertunç, Brain tumor classification using modified local binary patterns (LBP) feature extraction methods, *Med Hypotheses* **139** (2020), 109696.
- [61] J. Kang, Z. Ullah and J. Gwak, MRI-Based Brain Tumor Classification Using Ensemble of Deep Features and Machine Learning Classifiers, *Sensors* **21** (2021), 2222.
- [62] J. Amin, M. Sharif, M. Raza, T. Saba and A. Rehman, Brain Tumor Classification: Feature Fusion, In *Proceedings of the 2019 International Conference on Computer and Information Sciences (ICCIS)*, Sakaka, Saudi Arabia, 3–4 April 2019, pp. 1–6.
- [63] S.K. Baranwal, K. Jaiswal, K. Vaibhav, A. Kumar and R. Srikantaswamy, Performance analysis of Brain Tumour Image Classification using CNN and SVM, In *Proceedings of the 2020 Second International Conference on Inventive Research in Computing Applications (ICIRCA)*, Coimbatore, India, 15–17 July 2020, pp. 537–542.
- [64] A. Gumaie, M.M. Hassan, M.R. Hassan, A. Alelaiwi and G. Fortino, A Hybrid Feature Extraction Method With Regularized Extreme Learning Machine for Brain Tumor Classification, *IEEE Access* **7** (2019), 36266–36273.
- [65] A. Minz and C. Mahobiya, MR Image Classification Using Adaboost for Brain Tumor Type, In *Proceedings of the 2017*

- IEEE 7th International Advance Computing Conference (IACC), Hyderabad, India, 5–7 January 2017, pp. 701–705.
- [66] S. Gayathri, D.J.W. Wise, V. Janani, M. Eleaswari and S. Hema, Analyzing, Detecting and Automatic Classification of Different Stages of Brain Tumor Using Region Segmentation and Support Vector Machine, In *Proceedings of the 2020 International Conference on Electronics and Sustainable Communication Systems (ICESC)*, Coimbatore, India, 2–4 July 2020, pp. 404–408.
- [67] A. Sarkar, M. Maniruzzaman, M.S. Ahsan, M. Ahmad, M.I. Kadir and S.M.T. Islam, Identification and Classification of Brain Tumor from MRI with Feature Extraction by Support Vector Machine, In *Proceedings of the 2020 International Conference for Emerging Technology (INCET)*, Belgaum, India, 5–7 June 2020, pp. 1–4.
- [68] A.R. Mathew and P.B. Anto, Tumor detection and classification of MRI brain image using wavelet transform and SVM. In *Proceedings of the 2017 International Conference on Signal Processing and Communication (ICSPC)*, Coimbatore, India, 28–29 July 2017, pp. 75–78.
- [69] G. Cinarer and B.G. Emiroglu, Classification of Brain Tumors by Machine Learning Algorithms, In *Proceedings of the 2019 3rd International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMISIT)*, Ankara, Turkey, 11–13 October 2019, pp. 1–4.
- [70] R. Lavanyadevi, M. Machakowsalya, J. Nivethitha and A.N. Kumar, Brain tumor classification and segmentation in MRI images using PNN, In *Proceedings of the 2017 IEEE International Conference on Electrical, Instrumentation and Communication Engineering (ICEICE)*, Karur, India, 27–28 April 2017, pp. 1–6.
- [71] J. Amin, M. Sharif, M. Yasmin and S.L. Fernandes, A distinctive approach in brain tumor detection and classification using MRI, *Pattern Recognit Lett* **139** (2020), 118–127.
- [72] S. Prabha, R. Raghav, C. Moulya, K.G. Preethi and K. Sankaran, Fusion based Brain Tumor Classification using Multiscale Transform Methods, In *Proceedings of the 2020 International Conference on Communication and Signal Processing (ICCSP)*, Chennai, India, 28–30 July 2020, pp. 1390–1393.
- [73] V. Wasule and P. Sonar, Classification of brain MRI using SVM and KNN classifier, In *Proceedings of the 2017 Third International Conference on Sensing, Signal Processing and Security (ICSSS)*, Chennai, India, 4–5 May 2017, pp. 218–223.
- [74] J. Sachdeva, V. Kumar, I. Gupta, N. Khandelwal and C.K. Ahuja, A package-SFERCB-“Segmentation, feature extraction, reduction and classification analysis by both SVM and ANN for brain tumors”, *Appl Soft Comput* **47** (2016), 151–167.
- [75] K. Keerthana and S. Xavier, An Intelligent System for Early Assessment and Classification of Brain Tumor, In *Proceedings of the 2018 Second International Conference on Inventive Communication and Computational Technologies (ICICCT)*, Coimbatore, India, 20–21 April 2018; pp. 1265–1268.
- [76] B. Yin, C. Wang and F. Abza, New brain tumor classification method based on an improved version of whale optimization algorithm, *Biomed Signal Process Control* **56** (2020), 101728.
- [77] J. Cheng, Brain Tumor Dataset, 2017. Available online: https://figshare.com/articles/dataset/brain_tumor_dataset/1512427 (accessed on 2 June 2021).
- [78] E.S. Biratu, F. Schwenker, Y.M. Ayano, T.G. Debelee, A Survey of Brain Tumor Segmentation and Classification Algorithms, *Journal of Imaging* **7**(9) (2021), 179. <https://doi.org/10.3390/jimaging7090179>
- [79] Shreepad S. Sawant and Preeti S. Topannavar, Introduction to Probabilistic Neural Network – Used For Image Classifications, *International Journal of Advanced Research in Computer Science and Software Engineering* **5**(4) (2015), 279–283.
- [80] M. Akshay Kumaar, Duraimurugan Samiayya, Venkatesan Rajinikanth and P.M. Durai Raj Vincent, Seifedine Kadry (In Press), Brain Tumor Classification Using a Pre-Trained Auxiliary Classifying Style-Based Generative Adversarial Network, *International Journal of Interactive Multimedia and Artificial Intelligence*, vol. In Press, issue In Press, no. In Press, pp. 1–11. <https://doi.org/10.9781/ijimai.2023.02.008>
- [81] Maahi Amit Khemchandani, Shivajirao Manikrao Jadhav and B.R. Iyer, Brain Tumor Segmentation and Identification Using Particle Imperialist Deep Convolutional Neural Network in MRI Images, *International Journal of Interactive Multimedia and Artificial Intelligence* **7**(7) (2022), 38–47. <https://doi.org/10.9781/ijimai.2022.10.006>